Studies with stack of two different scintillators

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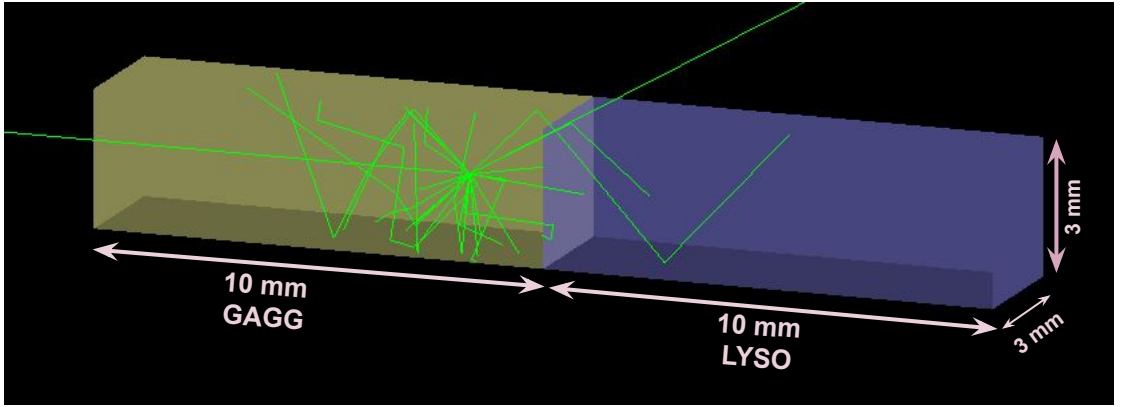
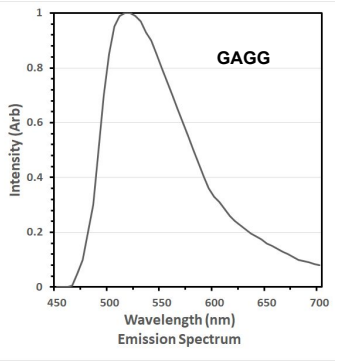
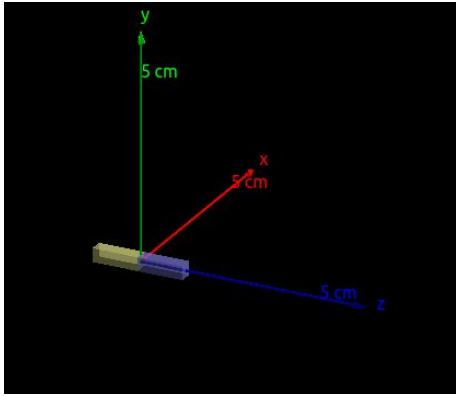
Geometry



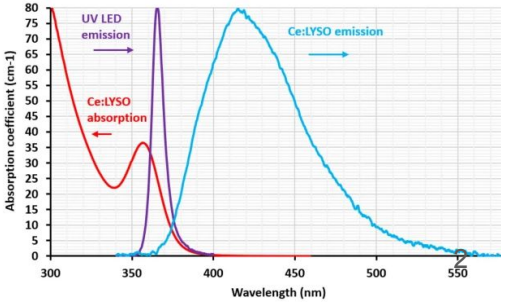
Yellow: GAGG (-10 < Z < 0)
Blue: LYSO (0 < Z < 10)

Scintillator to air surface: dielectric_metal
No additional surface defined between the scintillators.





Parameter	Material				
	PbWO4	BGO	LYSO	GAGG	LaBr3
Density (g/cm3)	8.28	7.13	7.1	6.63	5.29
SY (pho/MeV)	200	8200	30000	28244	61000
Rise time	10 ps	30 ps	70 ps	8 ns	5.7 ns
SY1	0.8	0.1	1	0.392	0.9
τ_1 (ns)	5	60	40	50.1	30
SY2	0.2	0.9	0	0.608	0.1
τ_2 (ns)	30	300	—	321.5	300
Peak emission λ(nm)	420 & 520	480	410	380 (weak) & 520	380
Birks Constant	0				



All Scintillation Parameters

No absorption length specified in the material.

One check

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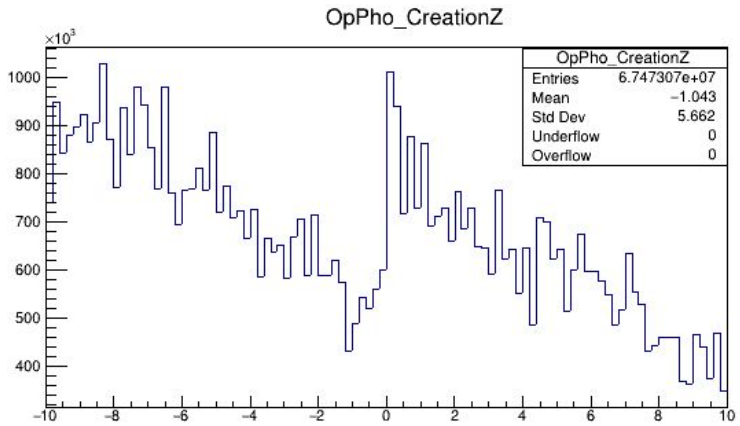
*******************************************************************************************************
* G4Track Information:   Particle = e-,   Track ID = 2,   Parent ID = 1
*******************************************************************************************************

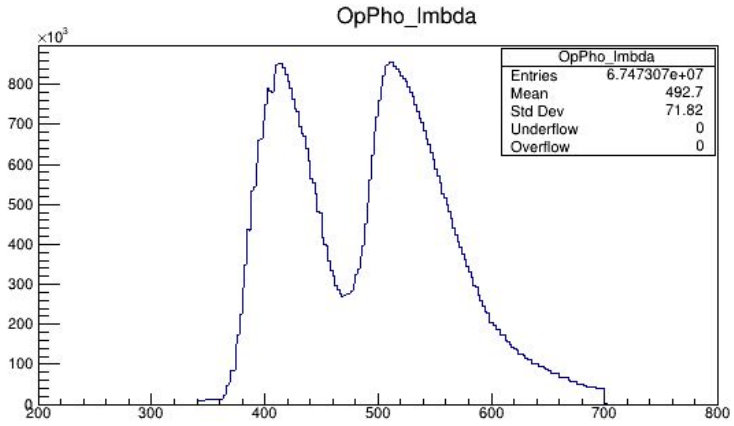
Step#    X(mm)    Y(mm)    Z(mm) KinE(MeV)  dE(MeV) StepLeng TrackLeng  NextVolume ProcName
   20  -0.0384  -0.0228    -2.87    0.0854        0        0     0.188     GAGG_PV initStep
   21  -0.0384  -0.0228    -2.87    0.0854        0 1.44e-05     0.188     GAGG_PV Cerenkov
   22  -0.0384  -0.0228    -2.87    0.0854        0 1.44e-05     0.188     GAGG_PV Cerenkov
   23  -0.0384  -0.0228    -2.87    0.0854        0 1.44e-05     0.188     GAGG_PV Cerenkov
   24  -0.0384  -0.0228    -2.87    0.0854        0 1.44e-05     0.188     GAGG_PV Cerenkov
   25  -0.0384  -0.0228    -2.87    0.0854        0 1.44e-05     0.188     GAGG_PV Cerenkov
   26  -0.0384  -0.0228    -2.87    0.0854        0 1.44e-05     0.189     GAGG_PV Cerenkov
   27  -0.0384  -0.0228    -2.87    0.0854        0 1.44e-05     0.189     GAGG_PV Cerenkov
   28  -0.0383  -0.0228    -2.87    0.0853 9.07e-05 1.44e-05     0.189     GAGG_PV Cerenkov
   29   -0.035  -0.0214    -2.87    0.0496   0.0357   0.0182     0.207     GAGG_PV eIoni
```

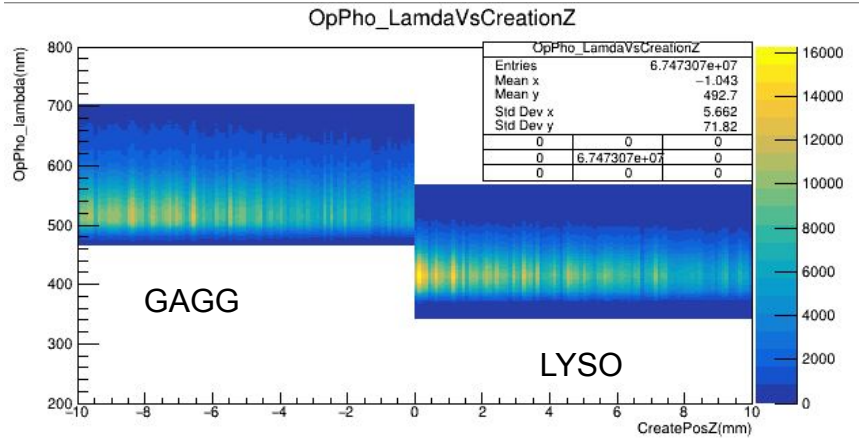
Don’t know why, but there is significant amount of Cerenkov produced in GAGG.
Same thing is observed in LYSO also.

For the studies presented in following slides, only scintillation photons are considered (and cerenkov photons are ignored).

Scintillation photons at the position of creation





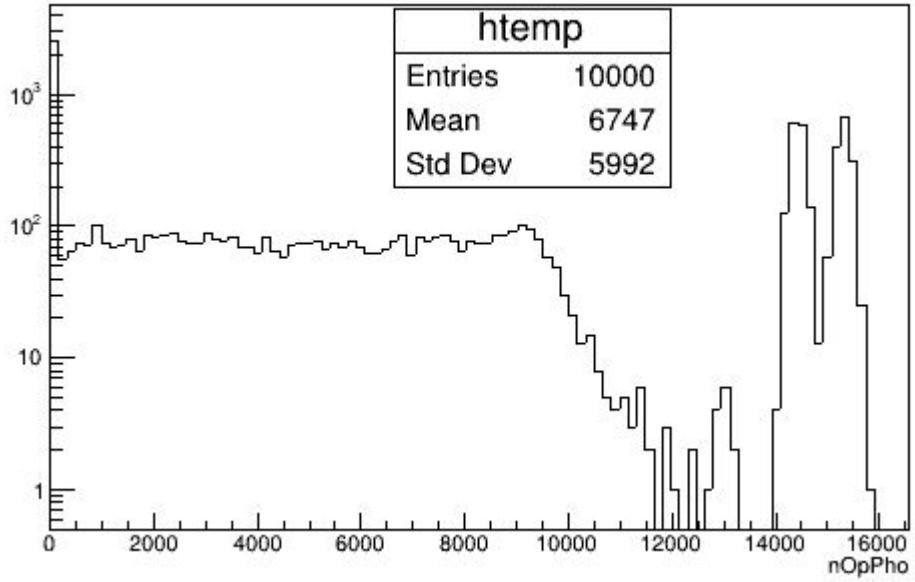


Filling the optical photon in the histogram only when it is at the point of creation (only one entry per optical photon).

```
if(particleName == "opticalphoton" && CreatorProc == "Scintillation"){
  if(preStepPoint->GetPosition() == CreatPos){
    fEventAction->nOpPhotons++;
    OpPho_Energy = track->GetKineticEnergy();
    OpPho_lmbda = (1239.8e-6)/OpPho_Energy;   //unit: nm

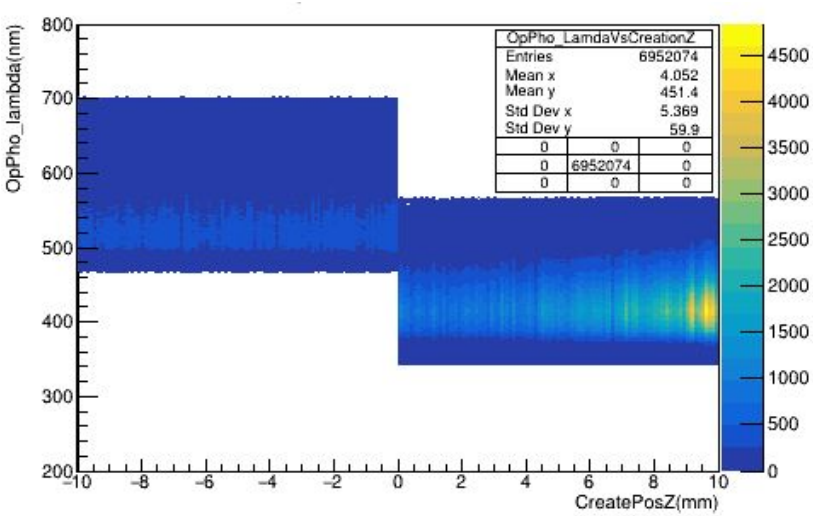
    G4AnalysisManager *man = G4AnalysisManager::Instance();
    man->FillNtupleDColumn(0, 1, CreatPos[2]);
    man->FillNtupleDColumn(0, 0, OpPho_lmbda);
    man->AddNtupleRow(0);
```

Distribution of number of optical photons in produced in an event



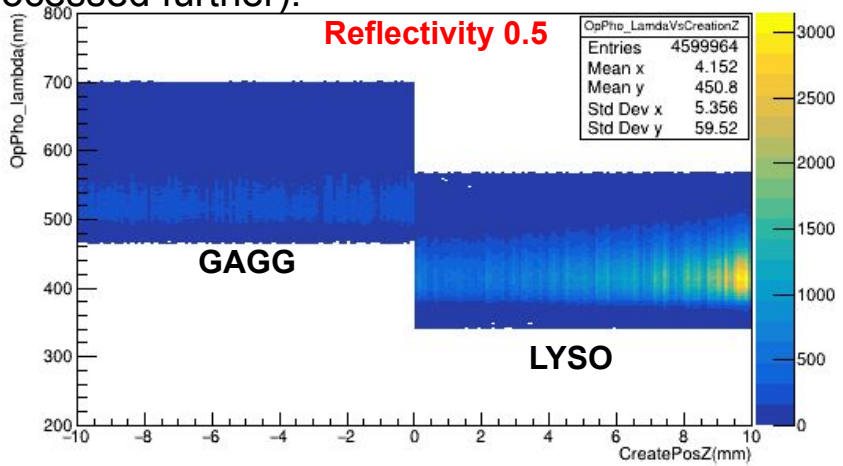
Two photopeaks might be corresponding to the different scintillation yield of the GAGG and LYSO.

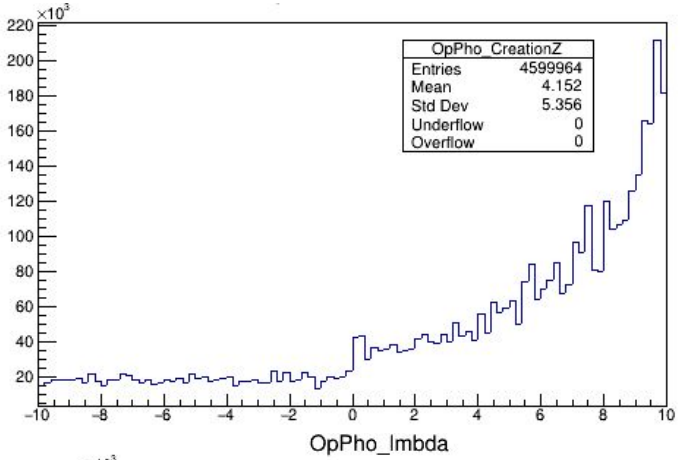
Optical photons at the rear surface (z = 10) with older reflectivity (0.5) and without killing the photons at the end



Optical photons at the rear surface (z = 10)

The optical photons are killed once they reach the rear surface after filling the histograms (i.e. not processed further).





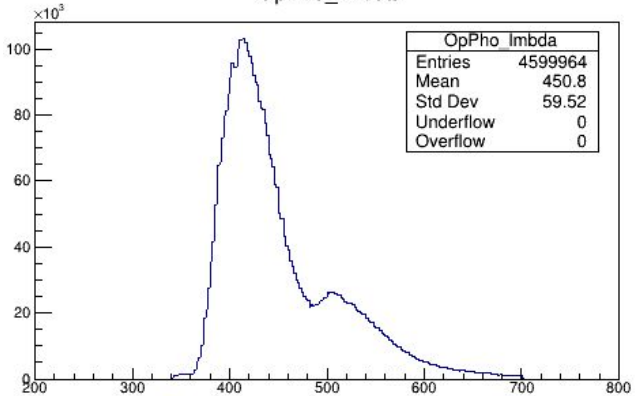

```
if(particleName == "opticalphoton" && CreatorProc == "Scintillation"){
  G4AnalysisManager *man = G4AnalysisManager::Instance();
  OpPho_Energy = track->GetKineticEnergy();
  OpPho_lmbda = (1239.8e-6)/OpPho_Energy;   //unit: nm

  if(preStepPoint->GetPosition() == CreatPos){
    fEventAction->nOpPhotons++;

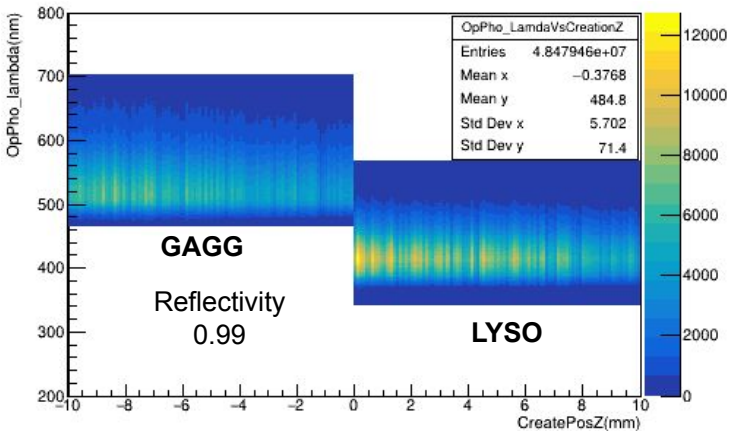
    man->FillNtupleDColumn(0, 1, CreatPos[2]);
    man->FillNtupleDColumn(0, 0, OpPho_lmbda);
    man->AddNtupleRow(0);
  }

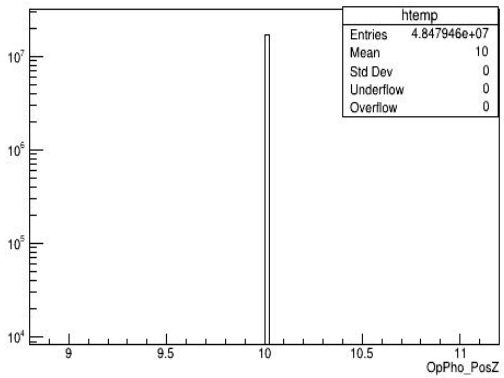
  if(OpPho_Pos[2] == 10*mm){
    man->FillNtupleDColumn(1, 1, CreatPos[2]);
    man->FillNtupleDColumn(1, 0, OpPho_lmbda);
    man->AddNtupleRow(1);

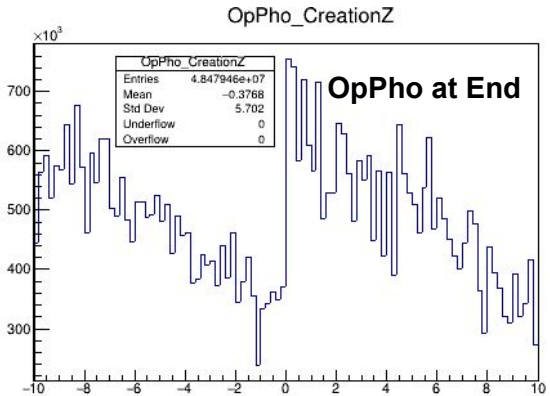
    track->SetTrackStatus(fStopAndKill);
```

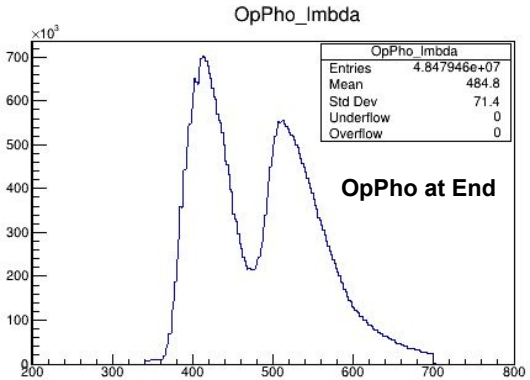


Increasing the reflectivity to 0.99







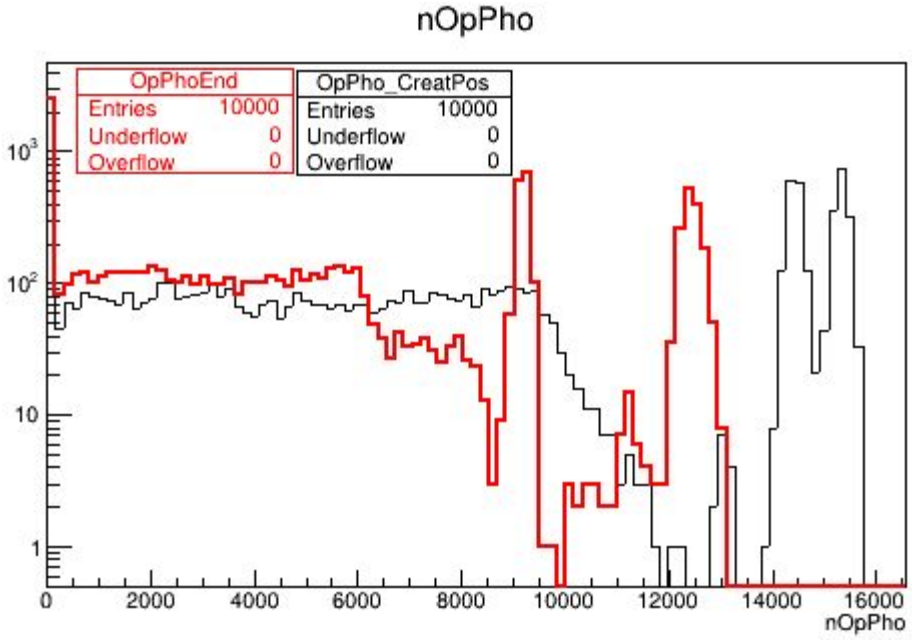


Optical photons reaching at the end are killed after filling their info in histogram (i.e. not processed further).

With higher reflectivity the asymmetry near z = 10 is gone (as compared to lower reflectivity), because with increased 0.99 reflectivity there are more chances that optical photon will reach to the detector end.

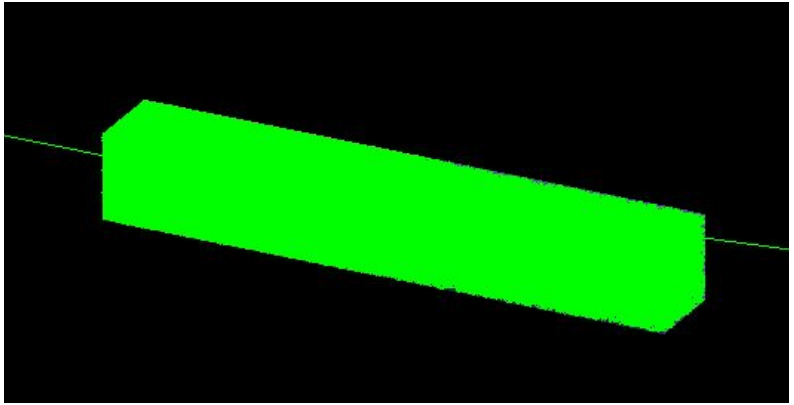
Still not sure why so many entries near z =0. Might be refractive index playing some role. We can check it by providing the same refractive index to both the materials (check the next slides).

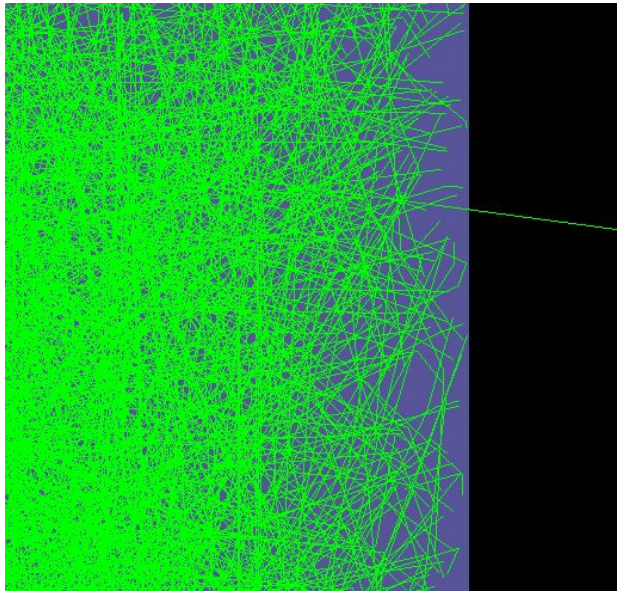
Distribution of nOptical Photons



Black - all scintillations photons created in the bulk
Red - only the scintillation photons reaching the rear surface of the tile stack (end of LYSO crystal).

Some event displays

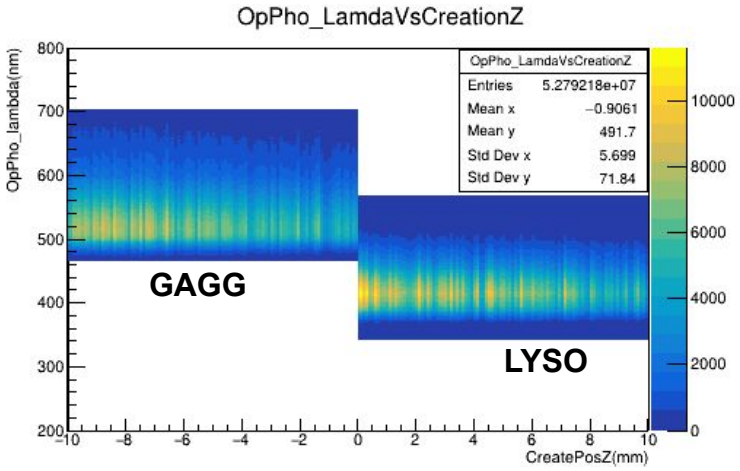




Rear surface of the tile stack

Some optical photons which are getting reflected the might be coming from cherenkov. Because killing only scintillation photons

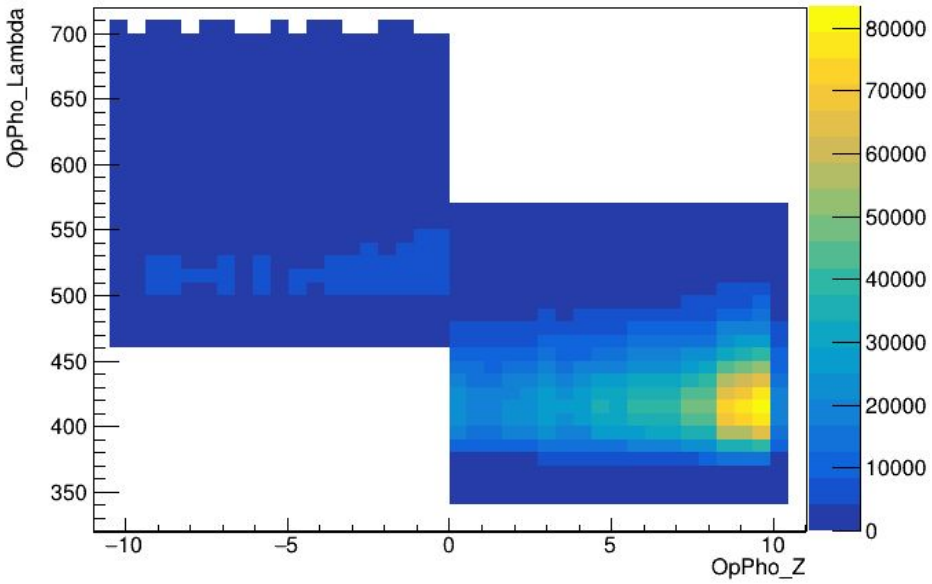
Checking with both the material same refractive index

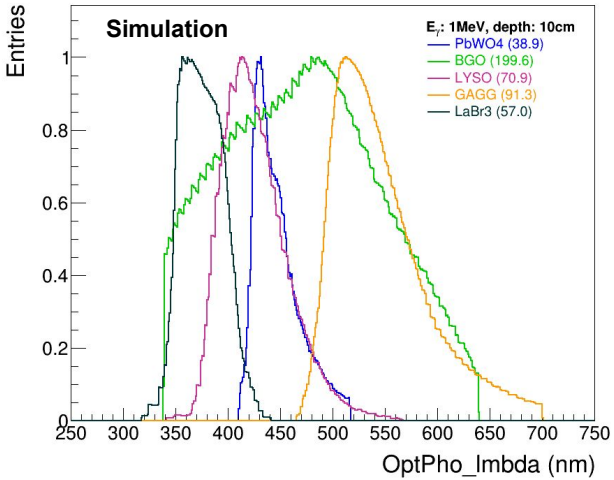


- Replaced the RI of LYSO with that of GAGG.
- Now there are considerably higher amount in GAGG also.
- This indicates that refractive index might be playing some role for more events near z =0 in the previous slide.

Backup & intermediate checks

Optical photon wavelength vs z of optical pho origin

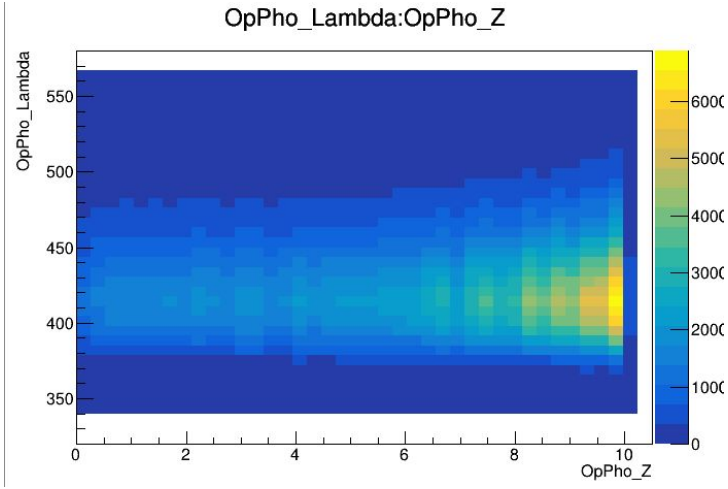




Here one thing makes sense that higher wavelength is coming from lower Z, which corresponds to gagg. And lower from LYSO (higher Z.)

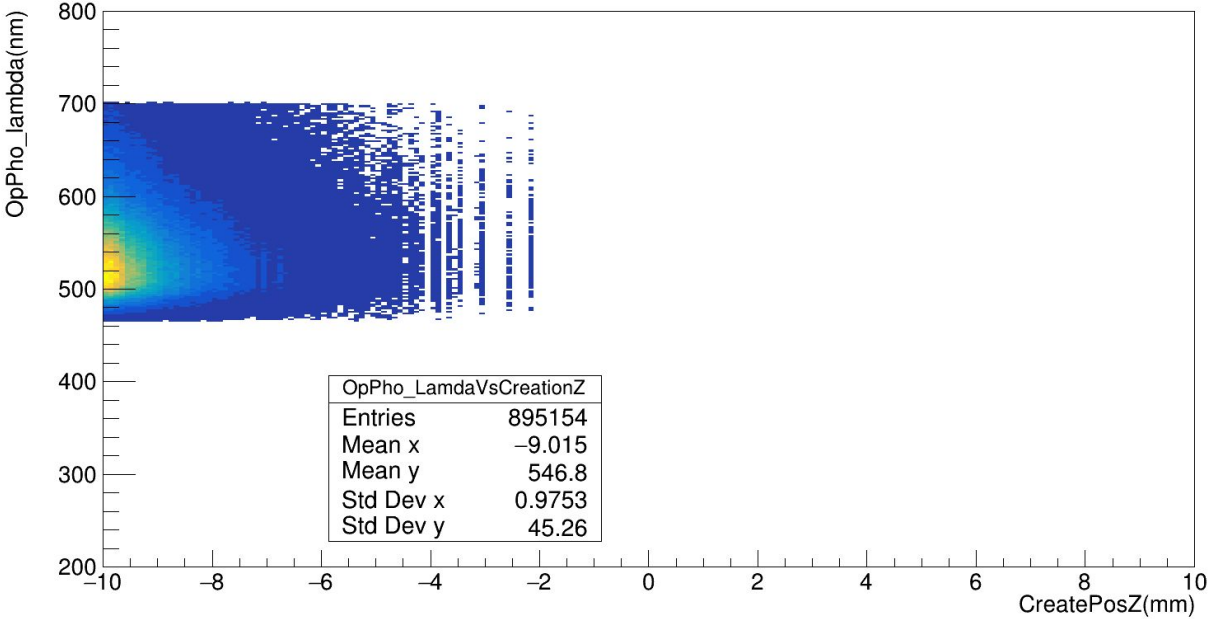
One thing not clear is why so much preference near Z = 10.

Checking why too many events near z = 10

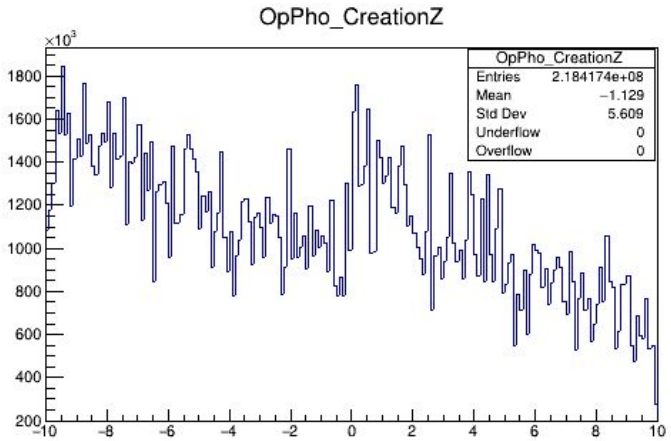


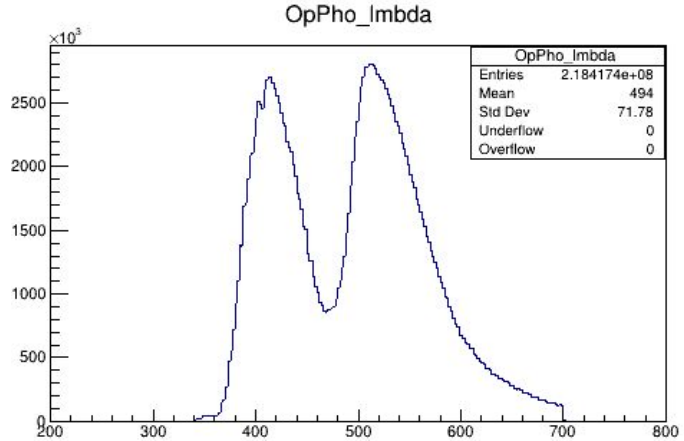
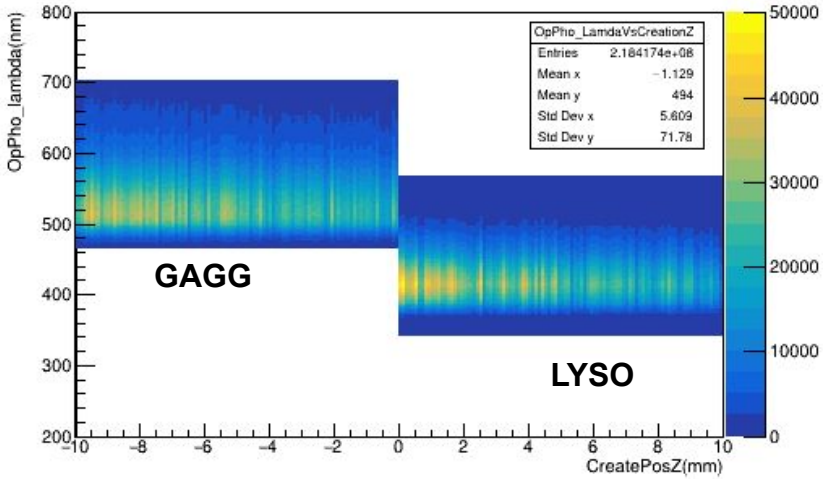
For lyso only

Shooting 100 keV photons



Checking after removing the z = 10 condition





- creation position of Z=10 for too many optical photons might be because of the condition that I ask for an optical photon to be at the end (z = 10).
- the reason could be that there is more probability of a optical photon to reach detector end if it is produced near the detector end, other wise the optical photon can get TIR from other surfaces and won't reach the end.
- I see more entries near z =0, those could be the optical photons which crossing from gagg to LYSO.
- **Here there are multiple entries for a single optical photons (each step point) as no condition on Z pos of optical photon.**